Flagging students at risk of failing, from the start of the year

Machine Learning × Advanced Python · final project report · PGE 2 · Shauryaman Singh & Chinmay Hajare · November 2026
Repository: github.com/sureyeahman/student-performance

1. Problem

User and decision. A school counsellor, at the start of the school year, has to decide which students to offer extra support (tutoring, a meeting with the family) before it is too late. Our tool predicts each student's final grade in Portuguese from information available on the first day, and flags the students likely to fail.

Target and task. The target is G3, the final grade on a 0–20 scale; a student passes with 10 or more. We treat it as **regression**: the predicted grade tells the counsellor how far a student is from the pass mark, and a decision threshold then turns it into a yes/no flag.

Cost of each mistake. A **missed** failing student (false negative) gets no help and may fail the year: this is the costly error. An **unnecessary** flag (false positive) costs a counsellor's time and a meeting, and may make a student feel labelled. We therefore accept extra flags in exchange for catching most failing students, and set the goal: **flag at least 75 % of the students who will fail**. For the grade itself, an error of 2 points is twice as bad as an error of 1, so we measure accuracy with the mean absolute error (MAE).

2. Data

We use the Student Performance dataset (UCI id 320; Cortez and Silva, 2008; CC BY 4.0), Portuguese course file: **649 students** from two Portuguese secondary schools (GP and MS), 30 input columns plus the three period grades G1, G2 and G3. Inputs describe the student (age, sex, study time, past failures, absences, going out, alcohol use, health), the family (parents' education and jobs, family size and relations) and the school (support, reason for choosing it).

- **Types.** 13 numeric columns (counts and ordered 1–5 scales, kept as numbers) and 17 text columns: 13 binary (yes/no, F/M, …) and 4 nominal (mother's and father's job, reason, guardian).
- **Missing values and duplicates.** None.
- **Special code.** 15 students have G3 = 0. All 15 have 0 absences, and their first-period grades were 4 to 11: a 0 here means the student did not sit the final exam, not a real grade. We remove them (634 students left); the tool predicts the grade of students who sit the exam.
- **Leakage.** G1 and G2 are correlated with G3 at 0.83 and 0.92, but they are only known during the year. Our counsellor uses the tool before them, so both are removed. The task becomes harder, but it is the honest set-up.

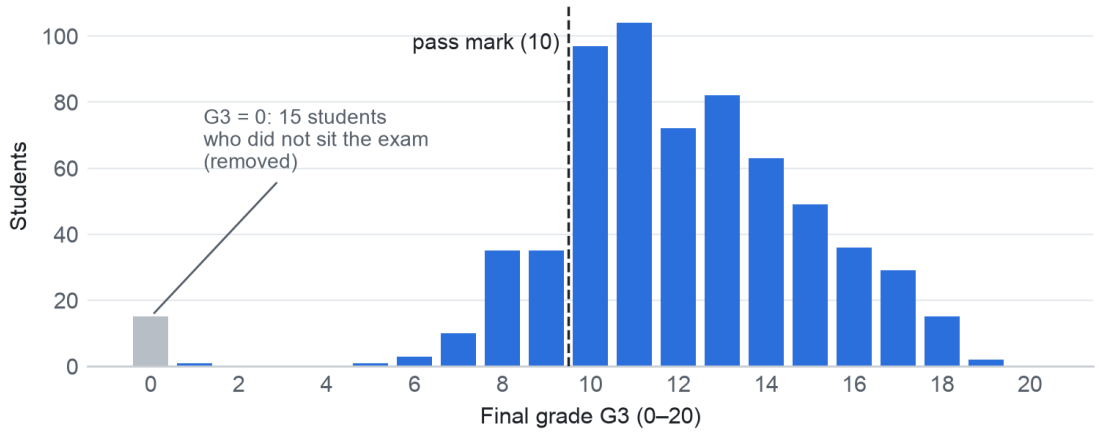


Figure 1. Final grades of the 649 students. After removing G3 = 0: mean 12.2, standard deviation 2.7, and 13.4 % of students fail.

Split. Once, before anything else: 80 % training (507 students) and 20 % test (127), **stratified on pass/fail** so both sets keep the same 13.4 % fail rate, with the fixed seed 42. The test set is not used again until the final evaluation.

3. Method

Pipeline. All preparation lives in one scikit-learn *Pipeline*: a *ColumnTransformer* that standardises the 13 numeric columns (*StandardScaler*) and one-hot encodes the 17 text columns (*OneHotEncoder*, one 0/1 column for binary variables, unknown categories ignored), followed by the model. The 30 columns become 43 features. Because the whole pipeline is passed to cross-validation, the scaler and encoder are refitted on the training folds only, and nothing leaks from the validation fold (one of our tests checks this).

Protocol. Every choice is made on the training set with the same 5-fold cross-validation (shuffled, seed 42): first two baselines, a *DummyRegressor* (predicts the mean) and a linear regression; then five models seen in class, each tuned with *GridSearchCV* on the same folds:

Model	How it predicts	Grid searched (config.yaml)
Ridge	linear; squared error + α Σ w²	α: 0.01, 0.1, 1, 10, 100, 300
Lasso	linear; squared error + α Σ |w| (can set weights to 0)	α: 0.001, 0.01, 0.03, 0.1, 0.3, 1
k-NN	average grade of the k most similar students	k: 3, 5, 10, 20, 40 × weights: uniform, distance
Linear SVR	linear; absolute error + ½ Σ w²	C: 0.001, 0.01, 0.1, 1
Huber	linear; Huber loss + α Σ w²	ε: 1.1, 1.35, 1.5, 2.0 × α: 0.0001, 0.01, 1, 10

Table 1. Candidate models and hyperparameter grids.

Metric and loss. Models are ranked by **MAE**: it is in grade points and easy to explain to a counsellor ("off by 1.7 points on average"). We also report RMSE, which weighs large misses more. As training loss we use the **Huber loss** for the final model: it is squared for small errors, like MSE, but grows only linearly for large ones, so a few students with an unexpected result (a sudden collapse, a surprise success) pull the model less than with MSE. Its cut-off ε and penalty α are chosen by cross-validation.

Decision threshold. The tool flags a student when the predicted grade is below a threshold. We choose it on **out-of-fold predictions** of the training set (*cross_val_predict*: each student is predicted by a model that never saw them) as the lowest threshold that reaches our 75 % recall goal, searched from 10 to 20 in steps of 0.1.

4. Results

Model (5-fold CV, training set)	Best hyperparameters	MAE	± std	RMSE
Huber (chosen)	ε = 2.0, α = 10	**1.825**	0.104	2.297
Ridge	α = 10	1.827	0.099	2.308
Lasso	α = 0.03	1.831	0.103	2.307
Linear regression (baseline)	—	1.847	0.098	2.331
Linear SVR	C = 1	1.887	0.117	2.356
k-NN	k = 10, distance	1.893	0.104	2.351
Dummy, mean (baseline)	—	2.255	0.114	2.737

Table 2. Cross-validated errors in grade points (lower is better). "± std": spread of the MAE across the 5 folds.

Every model beats the dummy baseline clearly (2.26 → about 1.83): background information does carry signal. Huber, Ridge and Lasso are **tied**: their differences (under 0.01 points) are far smaller than the fold-to-fold spread (about 0.10). The more flexible k-NN and the SVR do slightly worse. We keep **Huber regression**: lowest CV MAE by our rule, robust to unusual students, and linear, so it can be interpreted. Ridge would have been an equally valid choice.

Threshold (predicted G3 < t)	10.0	10.5	11.0	11.3	11.5	12.0
Recall: failing students flagged	32 %	41 %	66 %	**75 %**	79 %	84 %
Precision: flagged students who fail	47 %	41 %	42 %	**40 %**	36 %	29 %
Share of all students flagged	9 %	13 %	21 %	**25 %**	29 %	39 %

Table 3. The flag on out-of-fold predictions of the 507 training students (68 of them fail).

Flagging at the pass mark itself (10) would catch only a third of failing students, because predictions are pulled towards the average. The threshold **11.3** is the lowest that reaches 75 %, flagging a quarter of students.

Final test (used once). The chosen pipeline, refitted on the 507 training students, scores **MAE 1.66** and **RMSE 2.12** on the 127 test students, in line with cross-validation (1.83): the model is not overfitted, and the small gap is chance in a small test set. With the threshold 11.3:

Test set (127 students)	Flagged	Not flagged	Total
Fails (G3 < 10)	**11**	6	17
Passes	22	88	110

Table 4. Confusion matrix on the test set: recall 65 % (11 of 17), precision 33 %, 26 % of students flagged.

The recall (65 %) is below the 75 % reached in cross-validation: with only 17 failing students in the test set, each one moves it by 6 points, so this gap is expected rather than a sign of failure.

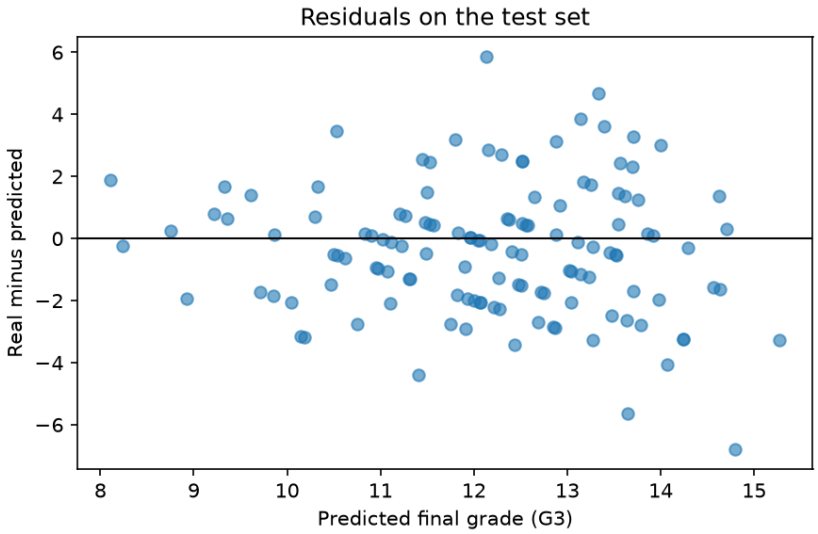


Figure 2. Residuals on the test set (real minus predicted grade).

Predictions stay between about 8 and 15 while real test grades range from 7 to 18: the model **over-predicts the weakest students** and **under-predicts the best**. The residuals are centred on 0 with no strong pattern, and a few students are missed by 4 to 7 points.

5. Interpretation

Huber is linear, so each coefficient is the change in predicted grade, all other features equal. Numeric features are standardised (change for one standard deviation); for yes/no features it is the gap between the two groups.

Feature	Coef.	Meaning
higher_yes	+1.15	Wants to go on to higher education: the strongest positive signal.
schoolsup_yes	−0.80	Gets extra school support. Not because support hurts: the school gives it to students who already struggle.
school_MS	−0.67	Attends the MS school rather than GP.
failures	−0.66	More past class failures, lower grade.
sex_M	−0.48	Boys score lower than girls in Portuguese, on average.
age	+0.39	Older students score slightly higher, other things (e.g. failures) equal.
romantic_yes	−0.38	In a romantic relationship.
Mjob_health	+0.38	Mother works in health care.
absences	−0.36	More absences, lower grade.
Medu	+0.33	More educated mother, higher grade.

Table 5. The ten largest coefficients of the final model.

Coefficients show **associations, not causes**: the model cannot say that wanting higher education raises a grade, only that the two go together.

6. Limits

- **Modest accuracy.** About 1.7 points of error on a 0–20 scale, and predictions squeezed towards the average. The tool ranks students by risk better than it predicts exact grades; this is why it is used through a threshold and not by reading the grade itself.
- **Small, local and old data.** 634 students from two Portuguese schools in 2005–2006. The model may not transfer to other schools, countries, subjects or years without being retrained and checked.

7. Risks of using it on real students

A flag is a prediction, not a judgement: on the test set, 2 in 3 flagged students in fact passed and 1 in 3 failing students was missed. It should only ever trigger an **offer of support**, never a sanction, a lower expectation or a decision made without a person, and a counsellor should always review it. The model uses sensitive information (sex, family situation, parents' jobs, alcohol use, relationships), so it can reproduce existing inequalities, for example by flagging boys or MS students more often; error rates should be checked group by group before any use. Collecting such data about minors requires a clear purpose, the consent of families, minimal data and secure storage under the GDPR.

Reproducing the results

`pip install -e ".[dev]"` · `pytest` · `student-performance train` · `student-performance evaluate` · `student-performance predict --input new_case.json`. All settings (paths, seed, folds, grids, recall goal) are in `config.yaml`; the notebook `notebooks/analysis.ipynb` tells the full story and calls the package.

Source. P. Cortez and A. Silva, *Using Data Mining to Predict Secondary School Student Performance*, FUBUTEC 2008, pp. 5–12. UCI Machine Learning Repository, dataset 320, CC BY 4.0.